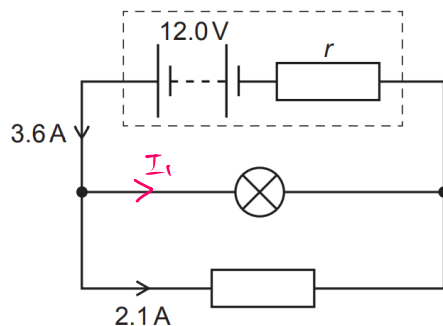


- 5 (a) (i) Since resistance is the ratio of potential difference to current, and from the graph, [M1]
From graph, this ratio is increasing and thus resistance increases. [A1]

(ii)



Current across filament lamp, $I_1 = 3.6 - 2.1 = 1.5 \text{ A}$ [1]
Reading off Fig. 4.2, p.d across filament lamp $V = 4.4 \text{ V}$ [1]

p.d across internal $r = 12.0 - 4.4 = 7.6 \text{ V}$
Internal resistance, $r = 7.6 / 3.6 = 2.11 \Omega$ [1]

(iii) $I = nAvq$ and current in series connection is constant

$$(nAvq)_{\text{filament}} = (nAvq)_{\text{copper}}$$

$$nAv_{\text{filament}}e = (2.5n)(360\text{A})(v_{\text{copper}})(e)$$

$$\text{ratio} = (2.5)(360) = 900$$
 [1]

(b) Potential difference across each lamp in series is 6 V, while potential difference across each lamp in parallel is 12 V. [1]

Since power = V^2 / R , each lamp in parallel receives more power than each lamp in series. Hence, temperature increases more for each lamp in parallel. [1]

Since resistance of filament lamp increases with temperature, resistance of each lamp in parallel will be greater. [1]

(c) From the first circuit, $V_{BC} = 1.5 \text{ V}$ and $V_{BD} = \frac{R_{BD}}{R_{BD} + 1.0} \times 2.0$ [1]

At balance length, $0.9375 \times V_{BD} = 1.5$

$$0.9375 \times \frac{R_{BD}}{R_{BD} + 1.0} \times 2.0 = 1.5$$

$$R_{BD} = 4.0 \Omega$$
 [1]

From the second circuit, to balance at point D, then

$$V_{BD} = 1.0 / 1.5 \times 1.5$$

$$\frac{4.0}{4.0 + 1.0 + R} \times 2.0 = 1.0$$
 [1]

Solving, $R = 3.0 \Omega$ [1]

$$6 \quad (2) \quad (i) \quad v = f\lambda \quad \Rightarrow \quad \lambda = \frac{v}{f} = \frac{320}{400} = 0.80 \text{ m} \quad [1]$$